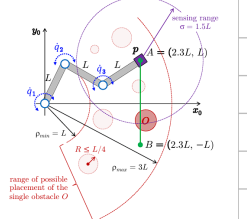
**Exercise #1**

Consider the planar 3R robot with links of equal length $L$ in Fig. 1, driven by joint velocity commands $\dot{\boldsymbol{q}} \in \mathbb{R}^3$. The end effector should trace the nominal linear path from $A$ to $B$ with a constant speed $v > 0$, while the entire robot avoids any collision with a single static obstacle $O$. The obstacle is of known circular shape but uncertain radius $R \leq L/4$, and is placed at least at a distance $\rho_{min} = L$ from the robot base and not farther than $\rho_{max} = 3L$. Its actual location is unknown a priori, but the obstacle can be detected by an omnidirectional proximity sensor mounted on the robot end effector and having a sensing range $\sigma = 1.5L$.

Define a sensor-based control scheme that makes the robot perform *at best* the assigned task, and describe qualitatively its expected performance.



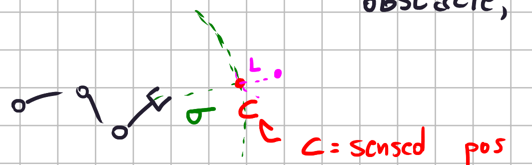

The path to perform is $p_d(t) = A + (B-A)vt \quad t \in [0, \frac{1}{v}]$

We always have to assume the worst case: the obstacle have a radius of $L/4$. The Jacobian is

$$J = L \begin{pmatrix} -s_1 - s_{12} - s_{123} & -s_{12} - s_{123} & -s_{123} \\ c_1 + c_{12} + c_{123} & c_{12} + c_{123} & c_{123} \end{pmatrix}$$

Until the obstacle is not perceived by the sensor, we can execute: $\dot{q} = J^{\#} \dot{p}_d + K_p(p_d - \xi)$ with $\xi(q)$ ee position.

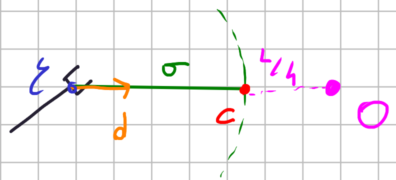
In the exact moment the obstacle is sensed, we know exactly is position, our circular range collides with the frontier of the obstacle, here we can estimate it's center:




$C = (c_x, c_y)$ = contact point

$D = C - \xi \quad d = \frac{D}{\|D\|}$ $d$ is the unit vector pointing from $\xi$ to $C$

$O$ = obstacle center $= \xi + d(\sigma + \frac{L}{4})$




From now on, we always have access to the position $O$, hence we can compute

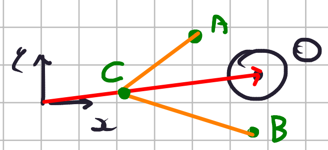
$$r_2 = \|\xi_2 - O\| \qquad r_3 = \|\xi_3 - O\| \qquad r_\xi = \|\xi - O\|$$

where $r_i$ = position of joint $i$. After knowing $O$, $A$ and $B$ i can compute a new path:

let $n = \frac{O}{\|O\|}$ i set $C = \frac{L}{2} n$ and i modify $p_d$:

$A' = \xi(q)$ in the exact moment where $O$ is percieved:

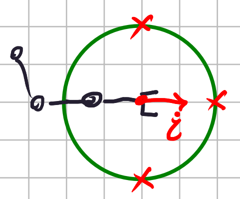
$$\hat{p}_d(t) = \begin{cases} A' + (C - A')2Tt & t \in [0, \frac{T}{2}] \\ C + (B - C)(\frac{2t}{T} - 1) & t \in [\frac{T}{2}, T] \end{cases}$$



note how $C$ is guaranteed to not be inside the obstacle.

We set $\dot{q} = J^{\#}\dot{\hat{p}}_d + (I - J^{\#}J)\nabla H$, $H = r_2 + r_3$

Note: assume that the sensor does not provide the position of the perceived frontier, this point can be defined any way by using the velocity of the end effector, in particular 3 possible contact points are found



in a conservative way we can assume there is an obstacle in each of these 3 points (if are coerent with the assumption).

**Exercise #2**

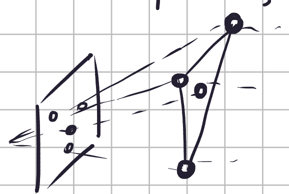
In an image-based visual servoing control scheme, the camera mounted on the robot end effector (eye-in-hand) looks for three specific points in the scene, characterized by their point features in the image plane with coordinates $(u_i, v_i)$, for $i = 1, 2, 3$. Derive the expression of the $2 \times 6$ *interaction matrix* $\boldsymbol{J}_b$ associated to the geometric barycenter $\boldsymbol{b} \in \mathbb{R}^2$ of the triangle in the image having these point features as the three vertices, namely

$$\dot{\boldsymbol{b}} = \boldsymbol{J}_b \begin{pmatrix} \boldsymbol{V} \\ \boldsymbol{\Omega} \end{pmatrix},$$

where $\boldsymbol{V} \in \mathbb{R}^3$ and $\boldsymbol{\Omega} \in \mathbb{R}^3$ are respectively the linear and angular velocity of the camera. Provide then an instantaneous motion of the camera such that $\dot{\boldsymbol{b}} = \boldsymbol{0}$.

the barycenter of the triangle is $(u, v)$ where $\begin{cases} u = \frac{1}{3}(u_1 + u_2 + u_3) \\ v = \frac{1}{3}(v_1 + v_2 + v_3) \end{cases}$

we denote $\hat{z}$ the estimated avg depth of these 3 points, that is $\hat{z} = \frac{1}{3}\sum_{i=1}^{3} z_i$



$$J_b = \begin{bmatrix} -\frac{\lambda}{\hat{z}} & 0 & \frac{u_1+u_2+u_3}{3\hat{z}} & \frac{(u_1+u_2+u_3)(v_1+v_2+v_3)}{9\lambda} & -\left(\lambda + \frac{(u_1+u_2+u_3)^2}{9\lambda}\right) & \frac{v_1+v_2+v_3}{3} \\ 0 & -\frac{\lambda}{\hat{z}} & \frac{v_1+v_2+v_3}{3\hat{z}} & \lambda + \frac{(v_1+v_2+v_3)^2}{9\lambda} & -\frac{(u_1+u_2+u_3)(v_1+v_2+v_3)}{9\lambda} & -\frac{u_1+u_2+u_2}{3} \end{bmatrix}$$

I set $(V \ \Omega) = \left( \frac{\hat{z}}{\lambda} \ -\frac{v_1+v_2+v_2}{u_1+u_2+u_3} \ 0 \ 0 \ 0 \ \frac{3}{v_1+v_2+v_3} \right) \Rightarrow \dot{b} = J_b \begin{pmatrix} V \\ \Omega \end{pmatrix} = 0.$

$\in NS(J_b)$

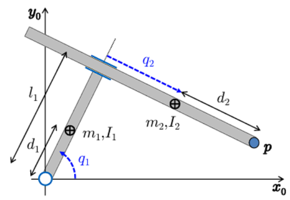
**Exercise #3**

With reference to the planar RP robot in Fig. 2, using the symbolic parameters specified therein, determine the complete expression of the *Cartesian inertia matrix* $\boldsymbol{M_p}(\boldsymbol{q})$ of the robot at the tip $\boldsymbol{p} \in \mathbb{R}^2$ as a function of the configuration $\boldsymbol{q}$.

Provide then the numerical value of $\boldsymbol{M_p}(\boldsymbol{q}^*)$ at $\boldsymbol{q}^* = \begin{pmatrix} 0 & 3 \end{pmatrix}^T$ [rad, m], using the parameters:

$l_1 = 1, \quad d_1 = 0.5, \quad m_1 = 3, \quad I_1 = 0.25; \qquad d_2 = 0.5, \quad m_2 = 0.5, \quad I_2 = 0.875.$

Assuming that this robot is at rest in $\boldsymbol{q}^*$ on a horizontal plane, check that the tip acceleration $\ddot{\boldsymbol{p}} \in \mathbb{R}^2$ has the same direction of any force $\boldsymbol{F} \in \mathbb{R}^2$ in the plane applied to the tip of the robot.




$$T_1 = \frac{1}{2}(I_1 + m_1 d_1^2)\dot q_1^2$$

$$p_{c2} = \begin{cases} l_1 c_1 + q_2 s_1 \\ l_1 s_1 - q_2 c_1 \end{cases} \Rightarrow \dot p_{c2} = \begin{cases} -l_1 s_1 \dot q_1 + \dot q_2 s_1 + q_2 c_1 \dot q_1 \\ l_1 c_1 \dot q_1 - \dot q_2 c_1 + q_2 s_1 \dot q_1 \end{cases} \Rightarrow T_2 = \frac{1}{2} m_2 (l_1^2 \dot q_1^2 + \dot q_2^2 + q_2^2 \dot q_1^2 - 2 l_1 \dot q_1 \dot q_2) + \frac{1}{2} I_2 \dot q_1^2$$

$$M_{11} = I_1 + m_1 d_1^2 + m_2(l_1^2 + q_2^2) + I_2 = e_1 + e_2 q_2^2$$

$$e_1 = I_1 + m_1 d_1^2 + m_2 l_1^2$$

$$M_{12} = -m_2 l_1 = e_3$$

$$M_{22} = m_2 = e_2$$

$$M(q) = \begin{pmatrix} e_1 + e_2 q_2^2 & e_3 \\ e_3 & e_2 \end{pmatrix} \qquad e_1 = 3/2 \qquad e_2 = 1/2 \qquad e_3 = -1$$

$$M_p = (J M^{-1} J^T)^{-1} \qquad r = \begin{cases} l_1 c_1 + (q_2 + d_2) s_1 \\ l_1 s_1 - (q_2 + d_2) c_1 \end{cases} \qquad J = \begin{pmatrix} -l_1 s_1 + (q_2 + d_2) c_1 & s_1 \\ l_1 c_1 + (q_2 + d_2) s_1 & -c_1 \end{pmatrix}$$

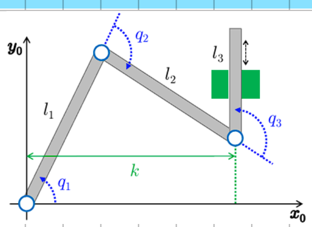
$$M(q^*) = \begin{pmatrix} 6 & -1/2 \\ -1/2 & 1/2 \end{pmatrix} \qquad J(q^*) = \begin{pmatrix} 3.5 & 0 \\ 1 & -1 \end{pmatrix} \Rightarrow M_p(q^*) = \begin{pmatrix} 0.445 & 0 \\ 0 & 1/2 \end{pmatrix}$$

$\Rightarrow \ddot q = M^{-1} F \Rightarrow$ not same dir.

**Exercise #4**

A 3R robot moves on a horizontal plane in the presence of geometric constraints in the Cartesian space, as illustrated in Fig. 3. Assume that the distance $k$ satisfies the inequalities $l_1 < k < l_1 + l_2$.

- Determine the dimension $M$ of the constraints and their possible expression $\boldsymbol{h}(\boldsymbol{q}) = \boldsymbol{0}$, with the associated Jacobian matrix $\boldsymbol{A}(\boldsymbol{q}) = \partial \boldsymbol{h}(\boldsymbol{q})/\partial \boldsymbol{q}$.
- Define the $(3-M) \times 3$ matrix $\boldsymbol{D}(\boldsymbol{q})$ to complete $\boldsymbol{A}(\boldsymbol{q})$ in a nonsingular way, as well as the blocks $\boldsymbol{E}(\boldsymbol{q})$ and $\boldsymbol{F}(\boldsymbol{q})$ of the inverse that appear in the *reduced dynamics* of this constrained robot. The reduced model should hold for any $\boldsymbol{q}$ such that the contact situation remains as in Fig. 3.
- Let $m_{ij}$ be the elements of the (symmetric) inertia matrix $\boldsymbol{M}(\boldsymbol{q})$ of this robot in free space. Give the full expression of the $(3-M) \times (3-M)$ *reduced inertia matrix* in the constrained case.




the contraint require two things: $r_{3x} = K$ and ee angle $= \frac{\pi}{2}$

$$r_{3x} = l_1 c_1 + l_2 c_{12} = K \Rightarrow h_1(q) = l_1 c_1 + l_2 c_{12} - K = 0$$

$$ee\,\alpha = q_1 + q_2 + q_3 \Rightarrow h_2(q) = q_1 + q_2 + q_3 - \frac{\pi}{2} = 0 \qquad \Rightarrow h = \begin{cases} h_1 \\ h_2 \end{cases}$$

$$A = \frac{\partial h}{\partial q} = \begin{pmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

i choose

$$D = \begin{pmatrix} -1 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} & 0 \\ 1 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} A \\ D \end{pmatrix}^{-1} = \frac{1}{l_1 s_1 + l_2 s_{12}} \begin{pmatrix} 1 & l_2 s_{12} & -l_2 s_{12} \\ -2 & l_1 s_1 - l_2 s_{12} & l_2 s_{12} - l_1 s_1 \\ 1 & l_2 s_{12} & l_1 s_1 \end{pmatrix}$$

$$\underbrace{\qquad\qquad}_{E} \qquad \underbrace{\qquad}_{G}$$

the reduced in. matrix is $G^T M G$ =

$$\begin{pmatrix} -l_2 s_{12} & l_2 s_{12} - l_1 s_1 & l_1 s_1 \end{pmatrix} \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix} \begin{pmatrix} -l_2 s_{12} \\ l_2 s_{12} - l_1 s_1 \\ l_1 s_1 \end{pmatrix} =$$

$$(-l_2 s_{12})(-l_2 s_{12} M_{11} + (l_2 s_{12} - l_1 s_1) M_{12} + l_1 s_1 M_{13}) +$$
$$(l_2 s_{12} - l_1 s_1)(-l_2 s_{12} M_{21} + (l_2 s_{12} - l_1 s_1) M_{22} + l_1 s_1 M_{23}) +$$
$$l_1 s_1 (-l_2 s_{12} M_{31} + (l_2 s_{12} - l_1 s_1) M_{32} + l_1 s_1 M_{33})$$

**Exercise #5**

Consider a 2R robot arm of human-like size and weight moving in a vertical plane with negligible friction. Two model-based controllers are being compared in a trajectory tracking problem in the joint space. The first is based on feedback linearization, yielding a control torque $\boldsymbol{u}_{FBL}(t)$, the second is a Lyapunov-based control law with global asymptotic convergence property, yielding a torque $\boldsymbol{u}_{GLB}(t)$.

Assuming that the PD gain matrices are the same in both control laws, provide the explicit expression of the torque difference $\boldsymbol{\Delta u}(t) = \boldsymbol{u}_{FBL}(t) - \boldsymbol{u}_{GLB}(t)$.

Next, when the desired joint trajectory and the PD gains are respectively

$$\boldsymbol{q}_d(t) = \begin{pmatrix} \frac{\pi}{2} + 3\sin\frac{\pi t}{2} \\ 1 - \cos 2\pi t \end{pmatrix}, \qquad \boldsymbol{K}_P = 100 \cdot I_{2\times2}, \quad \boldsymbol{K}_D = 20 \cdot I_{2\times2},$$

assume that at time $t = 2$ s, the current robot configuration is $\boldsymbol{q}(2) = \begin{pmatrix} \pi/2 & -\pi/2 \end{pmatrix}^T$ [rad] and the velocity tracking error is zero. For each robot joint, determine which is the controller that uses the larger instantaneous torque in absolute value.

the First is: $u_{FBL} = n(q, \dot{q}) + M(q)(K_P e + K_D \dot{e})$

the second is: $u_{GLB} = n(q_d, \dot{q}_d) + M(q_d)(\ddot{q}_d + K_P e + K_D \dot{e})$

$$\Delta u = n(q, \dot{q}) - n(q_d, \dot{q}_d) + (K_P e + K_D \dot{e})(M(q) - M(q_d)) - M(q_d)\ddot{q}_d$$

$q_d(2) = \begin{pmatrix} \frac{1}{2}\pi & 0 \end{pmatrix} \Rightarrow e = \begin{pmatrix} 0 & \pi/2 \end{pmatrix} \qquad \dot{e} = 0$

$K_P e = \begin{pmatrix} 0 & 50\pi \end{pmatrix} \qquad \dot{q}_d(2) = \dot{q}(2) = \begin{pmatrix} -\frac{3}{2}\pi & 0 \end{pmatrix}^T$

$M(q(2)) = \begin{pmatrix} e_1 & e_3 \\ e_3 & e_3 \end{pmatrix} \qquad n(q_d, \dot{q}_d) = \begin{cases} 0 \\ 0 \end{cases} \qquad \ddot{q}_d + K_P e = \begin{pmatrix} 0 \\ \pi(4\pi + 50) \end{pmatrix}$

$M(q_d(2)) = \begin{pmatrix} e_1 + 2e_2 & e_3 + e_2 \\ e_3 + e_2 & e_3 \end{pmatrix} \qquad n(q, \dot{q}) = \begin{cases} e_5 \\ e_5 + e_2 \frac{9}{4}\pi^2 \end{cases}$

$\ddot{q}_d(2) = \begin{pmatrix} 0 & 4\pi^2 \end{pmatrix}^T$

$\Rightarrow$

$u_{FBL} = \begin{cases} e_5 + e_3 50\pi \\ e_5 + e_2 \frac{9}{4}\pi^2 + e_3 50\pi \end{cases} \qquad u_{GBL} = \begin{cases} (e_3 + e_2)(4\pi^2 + 50\pi) \\ e_3(4\pi^2 + 50\pi) \end{cases}$

$u_{FBL,1} = g_0 m_2 d_{c2} + (I_2 + m_2 d_{c2}^2) 50\pi$

$\wedge$

$u_{GLB,1} = (I_2 + m_2 d_{c2}^2 + m_2 l_1 d_{c2})(4\pi^2 + 50\pi)$